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College Software Engineering Project | Confidential Draft for Academic Evaluation

SocialPulse

AI-Powered Social Media Analytics and Insights Platform

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Version 1.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of SocialPulse – AI-Powered Social Media Analytics and Insights Platform, a web-based application that enables content creators, businesses, digital marketers, and social media managers to analyze social media performance and receive actionable, data-driven recommendations from a single platform. The purpose of this document is to provide a clear, complete, and unambiguous description of the system's intended behaviour so that it can be used as the foundation for design, development, testing, and evaluation of the project by students and faculty.

1.2 Document Conventions

Functional requirements in this document are identified using the prefix FR-XX, non-functional requirements use the prefix NFR-XX, and use cases use the prefix UC-XX. Priority levels are classified as High (must-have for core functionality), Medium (important but not blocking), and Low (enhancement / future scope). Headings follow a numbered hierarchy consistent with IEEE 830 conventions.

1.3 Intended Audience and Reading Suggestions

This document is intended for the student development team, project guide/faculty evaluators, UI/UX designers, and reviewers assessing the project for academic submission. Developers should focus on Sections 3–5 for functional and architectural detail; evaluators may focus on Sections 1–2 and 6 for scope and quality expectations; designers should refer to Section 4 for interface requirements.

1.4 Project Scope

SocialPulse is a responsive, analytics-focused web application built to help users understand and improve social media performance rather than to publish or schedule content itself. The initial version (Phase 1) is a full-stack implementation in which data is supplied primarily through CSV/Excel dataset upload, with direct social media API integration supported as an optional data source where available. The system is architected so that live API synchronization, deeper generative-AI features, and additional platforms can be integrated in a later phase without redesigning the core system. The project consists of two modules:

- User Module — dataset/account import, KPI dashboard, post and content-type performance analysis, hashtag analysis, audience engagement and best-time insights, sentiment analysis, trend detection, AI-based recommendations, and report generation.
- Admin Module — management of user accounts, oversight of uploaded datasets and connected data sources, platform-wide statistics, and system configuration through an administrative dashboard.

Out of scope for this version: automated content publishing/scheduling, real payment or e-commerce integration, full-scale competitor/influencer monitoring, and multi-language sentiment analysis. These are listed as future enhancements in Section 2.7.

1.5 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
UI / UX	User Interface / User Experience
CRUD	Create, Read, Update, Delete
KPI	Key Performance Indicator
NLP	Natural Language Processing
ML	Machine Learning
Engagement Rate	(Likes + Comments + Shares) / Reach × 100, a core content-performance metric
Dataset Upload	CSV/Excel file containing social media post and engagement data, used as the primary data source in Phase 1
REST API	Representational State Transfer Application Programming Interface
Admin	A privileged user who manages platform accounts, data sources, and configuration
User / Creator	An end user who imports data and analyzes their own social media performance
Responsive Design	UI layout that adapts to desktop, tablet, and mobile screens

1.6 References

- IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications.
- Nielsen Norman Group — Usability Heuristics for User Interface Design.
- Scikit-learn and NLTK/spaCy documentation — used as technical reference for the analytics and NLP components.
- Platform developer documentation (e.g., Meta for Developers, X API) — referenced for optional future API integration.

2. Overall Description

2.1 Product Perspective

SocialPulse is a new, self-contained web product. It is not a modification of an existing system. Phase 1 operates as a client-server application: a React-based frontend communicates with a Python (Flask/FastAPI) backend over a REST API, with a database persisting user, account, post, sentiment, analytics, and recommendation data. The architecture uses a clear service-layer abstraction so that the primary dataset-upload data path can be extended with additional live social media API integrations later, allowing the system to evolve without major rework of UI components.

2.2 Product Features (Summary)

- Secure user registration, login, and profile management.
- Dataset upload (CSV/Excel) with validation, or optional social media account connection via API.
- Automated data cleaning and preprocessing.
- Interactive dashboard with KPI cards, growth charts, and trend indicators.
- Post-level performance ranking and content-type comparison.
- Hashtag frequency and performance analysis.
- Audience engagement pattern analysis and best-time-to-post recommendations.
- NLP-based sentiment analysis of audience comments.
- Trend detection (increasing / decreasing / stable) across key metrics.
- AI/ML-based recommendations for content type, hashtags, and posting schedule.
- Report generation and export (PDF/CSV/Excel).
- Admin dashboard for user, dataset, and platform-statistics oversight.
- Notifications, loading states, and meaningful empty/error states throughout the UI.

2.3 User Classes and Characteristics

User Class	Description	Technical Expertise
Content Creator / User	Imports data, analyzes performance, and reviews AI recommendations.	Basic — general web/app user
Business / Digital Marketer	Monitors campaigns, compares content performance, exports reports.	Basic–Medium
Administrator	Manages user accounts, data sources, and platform statistics.	Moderate — comfortable with admin panels
Guest Visitor	Views the public landing page without an account (limited actions).	Basic

2.4 Operating Environment

- Runs in modern web browsers: Chrome, Firefox, Edge, Safari (latest two versions).

- Responsive layout supporting desktop (≥1200px), tablet (≥768px), and mobile (≥320px) viewports.
- Client-server architecture in Phase 1 — a backend server and database are required to run the full application.
- Built using standard, widely available web and data-science technologies (see Section 5.2 Technology Stack).

2.5 Design and Implementation Constraints

- Phase 1 must remain fully usable via dataset upload alone; live social media API integration is optional and must not block core functionality.
- The system is constrained to open-source tools (React, Flask/FastAPI, Pandas, Scikit-learn, NLTK/spaCy) to avoid licensing costs.
- The codebase must be modular so that the data-import layer can later support additional platforms with minimal changes to components.
- The UI must remain understandable to non-technical users, favoring visual charts over dense raw tables.
- Project timeline and team size are constrained to a typical college semester project scope; features are intentionally kept realistic and achievable.
- All AI-generated recommendations must be clearly labeled as such and distinguished from directly computed analytics.

2.6 Assumptions and Dependencies

- Users have a device with a modern browser and internet access to load and use the application.
- Uploaded datasets follow a broadly consistent schema (post ID, date, engagement metrics, etc.), with minor variations tolerated by preprocessing.
- Sentiment analysis is performed primarily on English-language comments in Phase 1; multilingual support is a future enhancement.
- Dataset sizes for demonstration/academic purposes are moderate (hundreds to low thousands of records), not enterprise-scale big data.
- Availability of third-party social media APIs (where used) is subject to each platform's terms of service, rate limits, and approval process.
- Cloud hosting providers used for deployment remain available within their free/low-cost tiers for the academic project budget.

2.7 Future Enhancements (Post Phase 1)

- Live, real-time social media API synchronization across additional platforms.
- Advanced deep-learning models for sentiment classification and engagement prediction.
- AI-generated captions and content ideas.
- Automated content scheduling and publishing.
- Full-scale competitor and influencer monitoring.
- A natural-language analytics assistant (e.g., “Which post performed best this month?”).

- Multi-language sentiment analysis and reporting.

3. System Features and Functional Requirements

This section details the functional requirements of SocialPulse, organized by module and feature. Each feature includes a description, priority, and a set of testable functional requirements.

3.1 User Module

3.1.1 User Registration and Authentication

Description: Allows a user to create an account and log in securely to access their personal analytics workspace.

Priority: High

ID	Requirement
FR-01	The system shall allow a new user to register with name, email, and password.
FR-02	The system shall allow a registered user to log in and log out securely.
FR-03	The system shall validate email format and required fields before submission.
FR-04	The system shall allow a user to reset a forgotten password.
FR-05	The system shall persist the logged-in session using a secure token (JWT) until logout or expiry.

3.1.2 Dataset Import and Validation

Description: Enables users to import social media performance data via CSV/Excel upload or an optional API-based account connection.

Priority: High

ID	Requirement
FR-06	The system shall allow a user to upload a CSV/Excel dataset containing social media post data.
FR-07	The system shall validate uploaded datasets and report missing or invalid fields to the user.
FR-08	The system shall optionally allow a user to connect a social media account via API where supported.
FR-09	The system shall clean and preprocess imported data (duplicate removal, missing-value handling, text cleaning).
FR-10	The system shall display a clear error state when a dataset fails validation.

3.1.3 Dashboard and KPI Overview

Description: Provides a centralized, interactive dashboard summarizing account and content performance.

Priority: High

ID	Requirement
FR-11	The system shall compute KPIs including engagement rate, reach, impressions, and follower growth.
FR-12	The system shall display KPIs and growth trends on an interactive dashboard using charts and KPI cards.
FR-13	The system shall allow users to filter dashboard data by date range, platform, and content type.
FR-14	The system shall display an empty state when no data is available for the selected filters.

3.1.4 Post and Content-Type Performance Analysis

Description: Analyzes individual posts and compares performance across content types (image, video, text, etc.).

Priority: High

ID	Requirement
FR-15	The system shall rank posts by performance and identify top and low performers.
FR-16	The system shall display post-level detail including likes, comments, shares, views, reach, and engagement rate.
FR-17	The system shall compare average performance across content types.

3.1.5 Hashtag Analysis

Description: Analyzes hashtag usage frequency and the engagement associated with each hashtag.

Priority: Medium

ID	Requirement
FR-18	The system shall identify frequently used hashtags across a user's posts.
FR-19	The system shall calculate average engagement associated with each hashtag.
FR-20	The system shall highlight top-performing and low-performing hashtags.

3.1.6 Audience Engagement and Best-Time Analysis

Description: Analyzes when a user's audience is most active and recommends optimal posting windows.

Priority: High

ID	Requirement
FR-21	The system shall analyze engagement by day of week and hour of day.
FR-22	The system shall recommend suitable posting time windows based on historical engagement.
FR-23	The system shall present best-time recommendations in a simple, non-technical format (e.g., a heatmap or short statement).

3.1.7 Sentiment Analysis

Description: Classifies audience comments as positive, negative, or neutral using an NLP pipeline.

Priority: High

ID	Requirement
FR-24	The system shall perform sentiment analysis on comments and classify them as positive, negative, or neutral.
FR-25	The system shall display sentiment distribution (percentage breakdown) for a user's content.
FR-26	The system shall display sentiment trend over time.

3.1.8 Trend Detection

Description: Identifies whether key metrics are increasing, decreasing, or stable over a selected period.

Priority: Medium

ID	Requirement
FR-27	The system shall compare current-period metrics against a prior period.
FR-28	The system shall classify each tracked metric as increasing, decreasing, or stable.
FR-29	The system shall display trend direction with a clear visual indicator.

3.1.9 AI-Based Recommendations

Description: Generates actionable, data-backed recommendations for content strategy.

Priority: High

ID	Requirement
FR-30	The system shall generate recommendations for content type, hashtags, and posting time based on historical performance.
FR-31	The system shall clearly label AI-generated recommendations as distinct from directly computed analytics.
FR-32	The system shall optionally provide predictive analytics for expected engagement or reach.

3.1.10 Report Generation and Export

Description: Allows users to compile their analytics into a downloadable report.

Priority: High

ID	Requirement
FR-33	The system shall allow users to generate a report from their analytics.
FR-34	The system shall allow users to export reports in PDF, CSV, or Excel format.
FR-35	The system shall isolate each user's data from other users at all times.

3.1.11 Notifications and System Feedback

Description: Provides users with contextual feedback for actions and system states.

Priority: Medium

ID	Requirement
FR-36	The system shall show toast/notification messages for successful actions (e.g., “Dataset uploaded”).
FR-37	The system shall display loading indicators during data processing and analysis.
FR-38	The system shall display meaningful empty states (e.g., “No data yet — upload a dataset to get started”).
FR-39	The system shall display clear, non-technical error messages for invalid uploads or failed operations.

3.2 Admin Module

3.2.1 Admin Authentication

Description: Provides a separate, secured login for administrators.

Priority: High

ID	Requirement
FR-40	The system shall provide a distinct admin login screen.
FR-41	The system shall restrict access to admin routes to authenticated admin sessions only.

3.2.2 User Account Management

Description: Allows admins to oversee registered user accounts.

Priority: High

ID	Requirement
FR-42	The system shall allow the admin to view a list of all registered users.
FR-43	The system shall allow the admin to activate, deactivate, or delete a user account.
FR-44	The system shall list users in a searchable, tabular admin view.

3.2.3 Dataset and Data-Source Oversight

Description: Allows admins to monitor uploaded datasets and connected data sources across the platform.

Priority: Medium

ID	Requirement
FR-45	The system shall allow the admin to view a log of dataset uploads, including status (success/failed).
FR-46	The system shall allow the admin to view which social media platforms are connected across user accounts.

3.2.4 Platform Statistics Dashboard

Description: Gives admins a quick overview of platform-wide activity.

Priority: Medium

ID	Requirement
FR-47	The system shall display total counts of registered users, uploaded datasets, and analyzed posts.
FR-48	The system shall present platform statistics using summary cards and simple charts.

3.2.5 System Configuration Management

Description: Allows admins to adjust key analytical thresholds and settings.

Priority: Low

ID	Requirement
FR-49	The system shall allow the admin to configure the trend-detection threshold used to classify metrics as increasing/decreasing/stable.
FR-50	The system shall allow the admin to manage supported content-type categories used across the platform.

4. External Interface Requirements

4.1 User Interfaces

The application shall use a modern, clean, data-forward visual theme throughout, prioritizing clarity and readability of charts and KPIs over dense tabular data. Key screens include:

- Landing / Home page with product overview and sign-up/login call to action.
- Registration and Login pages.
- Data Upload page with CSV/Excel upload and validation feedback.
- Social Media Connection page (optional API-based account linking).
- Dashboard with KPI cards, growth charts, and quick insights.
- Post Analytics page with ranked, filterable post-performance list.
- Audience Analytics page with engagement heatmap and best-time recommendations.
- Sentiment Analysis page with distribution and trend charts.
- Trend Analysis page with metric trend indicators.
- Recommendations page with clearly labeled AI-generated suggestions.
- Reports page with report configuration and export controls.
- Profile / Settings page.
- Admin dashboard with statistics cards and management tables for users and datasets.

All screens shall be responsive and shall gracefully adapt layout, navigation (e.g., collapsing to a hamburger menu on mobile), and chart sizing across desktop, tablet, and mobile breakpoints. The UI shall include loading skeletons/spinners, toast notifications, and empty/error state illustrations with guidance text.

4.2 Hardware Interfaces

No special hardware is required. The system runs on any standard desktop, laptop, tablet, or smartphone capable of running a modern web browser.

4.3 Software Interfaces

Interface	Description
Backend REST API	Python Flask/FastAPI service used for authentication, data import, analytics, and reporting.
Database	PostgreSQL or MongoDB, used to persist user, account, post, sentiment, analytics, and recommendation data.
NLP Libraries	NLTK, spaCy, or Transformer-based models, used for sentiment classification.
Social Media Platform APIs	Optional data source for account-based import, subject to each platform's approval process.

4.4 Communication Interfaces

All client-server communication shall occur over HTTPS. The frontend and backend shall communicate via a documented REST API returning JSON payloads, following standard request/response conventions.

5. System Architecture and Data Design

5.1 High-Level Architecture

SocialPulse follows a layered client-server architecture designed for maintainability and future extension:

- Presentation Layer — Reusable React UI components (KPI cards, charts, tables, forms, modals) organized by feature (dashboard, posts, hashtags, sentiment, recommendations, admin).
- Application / API Layer — A Python Flask/FastAPI service exposing authentication, data-import, analytics, and reporting endpoints.
- Analytics & ML Layer — Pandas/NumPy for statistical computation, Scikit-learn for predictive models, and NLTK/spaCy/Transformers for sentiment analysis.
- Data Service Layer — An abstraction (e.g., datasetService, analyticsService, sentimentService) that currently processes uploaded datasets, but exposes function signatures that a future live-API integration would use in the same way.
- Persistence Layer — PostgreSQL or MongoDB storing user, account, post, sentiment, analytics, and recommendation data.

5.2 Technology Stack

Layer	Suggested Technology
Frontend Framework	React 18 + Vite (component-based, widely used for academic projects)
Styling	Tailwind CSS
State Management	React Context API or a lightweight state library
Backend Framework	Python — Flask or FastAPI, REST APIs
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
NLP	NLTK, spaCy, or Transformers (for sentiment analysis)
Database	PostgreSQL or MongoDB
Authentication	JWT-based authentication
Charts / Visualizations	Chart.js, Recharts, or Plotly
Deployment	Frontend: Vercel/Netlify · Backend: Render/Railway · Database: MongoDB Atlas / managed PostgreSQL

5.3 Core Data Entities

Entity	Key Attributes
User	id, name, email, passwordHash, createdDate, role (user/admin)
SocialMediaAccount	id, userId, platform, username, connectionInfo
Post	id, accountId, date, caption, contentType, hashtags[], likes, comments, shares, views, reach, impressions, engagementRate

Entity	Key Attributes
Sentiment	id, postId, commentText, sentiment, sentimentScore
Analytics	accountId, date, followers, engagement, reach, impressions, growthRate
Recommendation	id, userId, recType, recText, generatedDate

5.4 Sample Data Schema (JSON)

Example structure for a Post object as processed and stored in Phase 1:

```
{
  "id": "post_1042",
  "accountId": "acct_207",
  "date": "2026-07-18",
  "caption": "Behind the scenes of our new launch",
  "contentType": "video",
  "hashtags": ["launch", "bts", "product"],
  "likes": 842,
  "comments": 63,
  "shares": 27,
  "views": 15600,
  "reach": 12800,
  "impressions": 17300,
  "engagementRate": 7.29
}
```

Sample Post object showing engagement and content-type fields used across the analytics engine.

5.5 Database Table Design

The tables below summarize the primary persistence units and their purpose, mirroring the core data entities defined in Section 5.3.

Table / Collection	Purpose
users	Stores registered user accounts and authentication data.
social_media_accounts	Stores linked accounts (per user) used as an optional data source.
posts	Stores imported/processed post-level data for each account.
sentiments	Stores comment-level sentiment classification linked to posts.
analytics	Stores time-series KPI snapshots per account (followers, engagement, reach, growth).
recommendations	Stores AI-generated recommendations per user.

5.6 Future Backend / API Expansion Plan

The data-service layer is designed so each function (e.g., getDashboardOverview(), getRecommendations()) can be extended from a dataset-upload-driven implementation to include live API-sourced data with the same function signature. Planned REST endpoints for Phase 1 core functionality include:

Method	Endpoint	Purpose
POST	/api/auth/register	Register a new user account.
POST	/api/auth/login	Authenticate a user and issue a JWT.
POST	/api/data/upload	Upload and validate a CSV/Excel dataset.
GET	/api/dashboard/overview	Fetch aggregated KPI data for the dashboard.
GET	/api/posts	Fetch ranked post-performance data, with filters.
GET	/api/hashtags/performance	Fetch hashtag frequency and performance data.
GET	/api/sentiment	Fetch sentiment distribution and trend data.
GET	/api/recommendations	Fetch AI-generated recommendations for the current user.
POST	/api/reports/generate	Generate a report for a specified scope and date range.
GET	/api/admin/users	Admin: fetch all registered users.

6. Non-Functional Requirements

6.1 Performance

ID	Requirement
NFR-01	The dashboard shall load and render primary KPIs within 3 seconds for datasets up to 10,000 records.
NFR-02	Dataset upload and validation shall complete within 10 seconds for files up to 10 MB.

6.2 Usability

ID	Requirement
NFR-03	The interface shall follow a consistent, clean visual theme with clear visual hierarchy and legible typography.
NFR-04	Core tasks (upload dataset, view dashboard, generate report) shall be completable within 3 clicks/taps from the relevant screen.
NFR-05	The system shall provide clear feedback (notifications, states) for every user action.

6.3 Reliability and Availability

ID	Requirement
NFR-06	The system shall handle invalid files, missing values, and API failures gracefully without crashing.
NFR-07	The deployed system shall target at least 95% uptime during the academic evaluation period.

6.4 Security

ID	Requirement
NFR-08	Passwords shall be hashed using a strong, salted algorithm (e.g., bcrypt) and never stored or displayed in plaintext.
NFR-09	Admin routes/pages shall not be accessible without a valid, authenticated admin session.
NFR-10	A user shall never be able to view or export another user's data.

6.5 Maintainability

ID	Requirement
NFR-11	The codebase shall be organized into reusable, feature-based components with a clearly separated data-service and analytics layer.
NFR-12	AI-generated recommendations shall be clearly labeled and distinguishable from directly computed analytics.

6.6 Portability and Responsiveness

ID	Requirement
NFR-13	The application shall function correctly on the latest two versions of major browsers (Chrome, Firefox, Edge, Safari).
NFR-14	The layout shall adapt responsively across desktop, tablet, and mobile screen sizes.

6.7 Scalability (Design-Level)

ID	Requirement
NFR-15	The architecture shall allow additional analytics modules and social media platforms to be added without redesigning the core system.

7. Other Requirements

7.1 Error Handling and Empty States

Every list-based or data-driven screen (dashboard, posts, hashtags, sentiment, recommendations, admin tables) shall define a distinct empty state (no data yet) and error state (failed to load/process), each with a short, friendly message and, where applicable, a call-to-action (e.g., “Upload your first dataset”).

7.2 Data Migration Consideration

Because Phase 1 relies primarily on dataset upload, the design shall keep processed data shapes close to the planned future live-API contract so that a migration or synchronization utility can later reconcile uploaded data with live-API data with minimal transformation.

7.3 Academic/Project Constraints

The feature set defined in this document is intentionally scoped to be achievable by a small student team within a single semester, while remaining representative of a professional, real-world social media analytics product for demonstration and evaluation purposes.

8. Appendices

Appendix A: Use Case Summary

Use Case ID	Use Case	Actor
UC-01	Register / Log in	User
UC-02	Upload dataset / connect account	User
UC-03	View dashboard KPIs	User
UC-04	Analyze post and content-type performance	User
UC-05	Analyze hashtag performance	User
UC-06	View audience engagement and best posting time	User
UC-07	View sentiment analysis	User
UC-08	View trend analysis	User
UC-09	Receive AI-based recommendations	User
UC-10	Generate and export report	User
UC-11	Manage user accounts	Admin
UC-12	Oversee datasets and data sources	Admin
UC-13	View platform statistics	Admin

Appendix B: Glossary (Extended)

Term	Meaning
Engagement Rate	A calculated KPI: (Likes + Comments + Shares) / Reach × 100
Trend Direction	Classification of a metric as increasing, decreasing, or stable relative to a prior period
Empty State	UI shown when a list/section has no data yet
Data Service Layer	Abstraction layer isolating UI components from the underlying data source (upload-based or API-based)

Appendix C: Assumptions Recap

- This SRS covers Phase 1 (dataset-upload-driven, with optional API integration) unless explicitly noted as a future enhancement.
- Sentiment analysis is scoped to English-language comments in Phase 1.
- The document will be revised as the design evolves during development and review.

End of Document — SocialPulse Software Requirements Specification, Version 1.0